STATS 200A Discussion 3

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Example 3.1

$$A_n \uparrow A : A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots \Leftrightarrow \lim_{n\to\infty} \mu(A_n) = \mu(A) \quad A = \bigcup_{n=1}^{\infty} A_n$$

Solution.

$$\begin{aligned}
B_1 &= A_1 \\
B_2 &= A_2 \backslash A_1 \\
B_3 &= A_3 \backslash A_2 \\
&\vdots \\
B_i &= A_i \backslash A_{i-1} \\
\sum_{i=1}^{\infty} \mu(B_i) &= \mu\left(\bigcup_{i=1}^{\infty} B_i\right) \\
&= \mu(A_1) + \sum_{i=1}^{N} \mu\left(A_i \backslash A_{i-1}\right) \\
&= \mu(A_1) + \sum_{i=2}^{N} \left(\mu(A_i) - \mu(A_{i-1})\right) \\
&= \mu(A_1) + \left(\mu(A_n) - \mu(A_1)\right) \\
&= \mu(A_n) \\
\sum_{i=1}^{\infty} \mu(B_i) &= \lim_{n\to\infty} \mu(A_n)
\end{aligned}$$

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Example 3.2

$$A_n \downarrow A : A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots \Leftrightarrow \lim_{n\to\infty} \mu(A_n) = \mu(A), \quad \mu(A_1) < \infty$$

Solution.

$$\begin{aligned} B_1 &= A_1 \backslash A_2 \\ B_2 &= A_2 \backslash A_3 \\ &\vdots \\ B_i &= A_i \backslash A_{i+1} \end{aligned}$$

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Remark 3.3

$$\Omega, \mathcal{F} = \begin{cases} \emptyset, \Omega \in \mathcal{F} \\ A \in \mathcal{F} \Leftrightarrow A^c \in \mathcal{F} \\ A_1, A_2, \ldots, \mathcal{F} \to \bigcup_{i=1}^{\infty} A_i \in \mathcal{F} \end{cases}$$

$$\mu_{\mathbb{P}} : \mathcal{F} \to [0,1] \begin{cases} \mu(\emptyset) = 0 \\ A_1, A_2, \ldots \text{pairwise disjoint} \quad \mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) \end{cases}$$

$$X : \Omega \to \mathbb{R}$$

Example 3.4

Ω: roll 2 dice

$$\Omega = \begin{Bmatrix} (1,1) & (1,2) & \cdots & (1,6) \\ (2,1) & \ddots & & \\ \vdots & & \ddots & \\ (6,1) & & & (6,6) \end{Bmatrix}$$

X = sum of 2 dice $\in \{2,3,4,\ldots,12\} = \mathbb{X}$
$X((2,1)) = 3, X((3,4)) = 7$
$A \subseteq \mathbf{X}$
$X^{-1}(A) := \{\omega : \Omega \mid X(\omega) \in A\}$
$A = \{2,3,4\}$
$X^{-1}(A) = \{(1,1),(2,1),(1,2),(3,1),(2,2),(1,3)\}$
$X^{-1}(\{2\}) = \{(1,1)\}$
$X^{-1}(\{3\}) = \{(1,2),(2,1)\}$
$X^{-1}(\{4\}) = \{(1,3),(2,2),(3,1)\}$
$X^{-1}(\{5\}) = \{(1,4),(2,3),(3,2),(4,1)\}$
$\sigma(X) = X^{-1}(\sigma) = \sigma\left(\left\{B \subseteq \Omega : \exists A \in y \,\text{st}\, B = X^{-1}(A)\right\}\right)$
$Z = \mathbf{1}_{X\%2=0} \in \{0,1\}$
$\mathcal{Z} = 2^{\{0,1\}} = \{\{0\},\{1\},\{0,1\},\emptyset\}$
$Z^{-1}(\{0\}) = \{(2,1),(1,2),(4,1),(4,1),(6,1),(1,6),(3,4),(4,3),(3,6),(6,3),(5,6),(6,5),(3,2),(2,3),\ldots\} = Z_0$
$Z^{-1}(\{1\}) = \{(1,1),(2,2),(2,4),(1,3),\ldots\} = \Omega \backslash Z^{-1}(\{0\}) = Z_1$
$\sigma(Z) = \{\emptyset, \Omega, Z_1, Z_2\}$